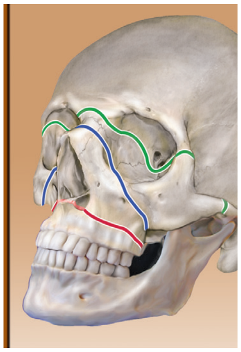

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Renal Medicine and Genitourinary Trauma in the Athlete

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Trauma to the genitourinary (GU) tract is relatively uncommon because of the anatomic location of key internal GU organs, the kidneys and bladder. However, prompt recognition of the signs and symptoms associated with GU trauma will allow the clinician to order appropriate imaging tests and implement therapeutic plans that can save organs and even a person's life.

DEFINITION (CLASSIFICATIONS)

Table 19.1 summarizes the classification of kidney trauma injuries according to severity. This classification correlates with the need for surgical intervention.[1]

EPIDEMIOLOGY

Of all the GU organs, the external genitalia (penis, scrotum, and vulva) are most likely to be injured. Of the internal GU organs, the kidneys are the most likely to be injured after a patient experiences abdominal trauma.[2]

Overall, the most common sport causing abdominal injury is cycling. Football, rugby, gymnastics, horseback riding, wrestling, martial arts, and hockey are a few of the sports that include activities capable of causing significant abdominal and consequent GU organ injury; the use of exercise equipment can do so as well.[3,4]

The overall incidence of renal injury as a result of trauma ranges from 1.4% to 3.25%.[5] Sports-related trauma to the kidney is uncommon and is reported to constitute only 15% to 20% of all traumatic renal injuries.[6] Most cases of renal trauma are the result of blunt trauma specifically relating to motor vehicle accidents and falls.[5] Kidney injuries are particularly common when a person is subjected to rapid deceleration forces. One analysis shows that of the 23,666 sports-related injuries among high school–age varsity athletes, only 18 kidney injuries were reported, none of which was serious.[4] In an analysis of more than 653,000 trauma cases from the National Trauma Data Bank, 16,585 were identified as trauma from bicycle injuries.[7] Only 2% of the patients in these cases experienced a GU tract injury, with the kidneys being the organ most likely affected (in 75% of cases), followed by the bladder and urethra (in 15% of cases) and the penis and scrotum (in 10% of cases). Sixty percent of the patients with GU injuries had evidence of concomitant fractures of the spine or pelvis, suggesting that isolated GU trauma is uncommon.[7] Compared with renal injuries, testicular injuries in sports occur at a much lower rate. A review of a trauma registry of all cases of renal and testes injuries (1.4% of all injuries) showed that 92% involved the kidneys and 8% the testes.[8] It is estimated that more than half of injuries to the testes occur during sporting events.[9]

PATHOBIOLOGY/PATHOPHYSIOLOGY

The kidneys are located in the retroperitoneal space and are surrounded by visceral fat and the Gerota fascia. The kidneys lie on either side of the spinal column in front of the psoas muscle and medial to the quadratus lumborum muscle. The hepatic flexure of the colon on the right and the spleen and the splenic flexure on the left cover the kidneys anteriorly. Because the kidneys are protected by surrounding structures, traumatic kidney injuries during sporting activities occur mainly in association with major forces and are usually associated with injury to other organs.

Injuries to the renal parenchyma constitute the vast majority of cases. Preexisting renal abnormalities such as hydronephrosis, renal cysts, or an abnormal renal anatomic position increase the likelihood of renal injury during trauma and are reported in 4% to 19% of adults and 12% to 35% of children.[5,10–12] These subjects have more severe symptoms and are more likely to require surgical intervention.[10–12] Vascular injuries of the kidneys occur during deceleration forces and result from damage to the renal pedicle. These cases may present with thrombosis or rupture of the vasculature.

Bladder injuries also occur as a consequence of blunt force trauma to the abdomen. The bladder's anatomic location deep in the anterior bony pelvis makes it less frequently injured by trauma. However, the weakest part of the bladder is the dome, which is mobile and susceptible to injury when the bladder is full.

The testes are particularly vulnerable to trauma because of their external location and lack of anatomic protection when blunt trauma forces the scrotum against the pelvic bone. Testicular rupture, scrotal wall hematoma, or intrascrotal hematocele are possible.[13]

DIAGNOSIS

Obtaining a thorough history is imperative. The initial evaluation of patients should include attention to vital signs recorded on the field and upon arrival at the hospital. The lowest recorded

Abstract

Trauma to the genitourinary (GU) tract is relatively uncommon because of the anatomic location of key internal GU organs, the kidneys and bladder. However, prompt recognition of the signs and symptoms associated with GU trauma will allow the clinician to order appropriate imaging tests and implement therapeutic plans that can save organs and even a person's life. Blunt trauma, often in the context of rapid deceleration, is the usual cause of kidney and bladder injury. Due the location of the kidneys, their injury is usually accompanied by simultaneous injury to other abdominal organs or bone fractures (spine and ribs). Hematuria is the most common presenting feature and should trigger evaluation of the GU tract. Although any hematuria should raise concern, gross hematuria is of particular concern. The presence of ascites in the context of abdominal trauma may be due to bladder perforation and associated urinary ascites. GU trauma may also be associated with renal metabolic complications including acute kidney injury, hyponatremia, and rhabdomyolysis. Although these are infrequent, they may be life threatening and should be considered given the appropriate presentation.

Keywords

ascites
bladder injury
genitourinary tract trauma
hematuria
kidney injury
kidney transplant
solitary kidney

TABLE 19.1 Organ Injury Severity Scale for the Kidney (American Association for the Surgery of Trauma)

Grade	Type	Description
I	Contusion	Microscopic or gross hematuria; urologic studies normal
	Hematoma	Subcapsular and nonexpanding without parenchymal laceration
II	Hematoma	Nonexpanding perirenal hematoma confined to the renal retroperitoneum
	Laceration	Less than 1 cm parenchymal depth of renal cortex without urinary extravasation
III	Laceration	Greater than 1 cm parenchymal depth of renal cortex without collecting system rupture or urinary extravasation
IV	Laceration	Parenchymal laceration extending through the renal cortex, medulla, and collecting system
	Vascular	Main renal artery or vein injury with contained hemorrhage
V	Laceration	Completely shattered kidney
	Vascular	Avulsion of the renal hilum, devascularizing the kidney

From Santucci RA, McAninch JW, Safir M, et al. Validation of the American Association for the Surgery of Trauma organ injury severity scale for the kidney. *J Trauma.* 2001;50:195–200.

systolic blood pressure may indicate the need for radiologic assessment of subjects for a kidney injury. Careful examination of the abdomen, chest, and back is critical. Patients with evidence of abdominal or flank tenderness or hematoma, rib fractures, and penetrating injuries to the low thorax or flank may have sustained an injury to the kidney and require further assessment. Pelvic fractures in trauma may alert the physician to the potential of bladder injury. Increasing abdominal girth with “ascites” without hemodynamic instability and without a drop in hemoglobin levels should be cause for suspicion, as the diagnosis can be delayed.[14] Persons with a testicular injury usually present with swelling, tenderness, and ecchymosis. Rupture of the testis is associated with immediate severe pain.[13]

Laboratory Findings

Hematuria, either microscopic or gross, is the best indicator of injury to the urinary tract after trauma. Although hematuria is seen in 80% to 90% of cases of kidney trauma, lack of hematuria does not eliminate the possibility; therefore a high degree of clinical suspicion should be maintained if the mechanism of injury suggests renal trauma. In addition, the degree of hematuria may not correlate with the degree of injury. However, in general, gross hematuria associated with blunt trauma increases the likelihood of major injury.

A rising serum creatinine in the absence of anuria/oliguria, especially in the context of ascites, could indicate bladder injury with intraperitoneal/intraabdominal urinary leak, as urinary ascites is resorbed across the peritoneum.

Imaging

Imaging studies specifically focused on the GU tract are required for all patients with rapid deceleration as the mechanism of blunt trauma (e.g., a motor vehicle accident or fall from a height), patients with hypotension, adults with gross hematuria, and children with microscopic hematuria.[15] Hemodynamically stable patients with only microscopic hematuria may not require further imaging but should undergo a thorough follow-up evaluation for potentially harmful delayed effects of trauma.[16]

An abdominal computed tomography (CT) scan with use of intravenous contrast is the imaging modality of choice in patients with trauma to the GU tract. In one series, the most common findings on CT were perirenal hematoma (29.4%), intrarenal hematoma (24.7%), and parenchymal disruption (17.6%).[17] Measurement of serum electrolytes and serum creatinine is useful in guiding diagnostic and treatment plans. Contrast should be avoided in those with severely reduced renal function, although emergent situations may necessitate its use.

For suspected bladder injury, CT and plain retrograde cystography are equivalent imaging modalities that would demonstrate extravasation of contrast.[18] Early diagnosis is essential for testicular salvage when there is trauma to the scrotum, the likelihood of which decreases with time. Scrotal ultrasonography is a safe, noninvasive, and valuable tool for rapid detection of testicular rupture, hematocele, hematoma, or traumatic torsion.[13]

Differential Diagnosis

Exercise-induced hematuria is a relatively common, benign finding among athletes. The incidence ranges between 50% and 80%, with the highest incidence reported among swimmers, track athletes, and lacrosse players.[19] A thorough medication history is critical, particularly for use of nonsteroidal inflammatory drugs (NSAIDs). In one study, more than half of the athletes with idiopathic hematuria regularly used NSAIDs.[20] Preexisting glomerular or cystic kidney disease may be the source of microscopic hematuria and must be differentiated from trauma-induced hematuria. The presence of proteinuria may suggest a preexisting glomerular lesion.

TREATMENT

A detailed discussion of specific surgical management is beyond the scope of this chapter. However, awareness of the following therapeutic considerations is important:

1. Surgical exploration of the kidney is required in cases with severe, persistent, or life-threatening renal hemorrhage, pedicle avulsion, or an expanding retroperitoneal hematoma.[5]
2. A nephrectomy may be required in persons who have sustained severe trauma and in unstable patients. However, patients who are more stable, especially those with a solitary kidney or with bilateral injuries, may be candidates for renovascular repair and reconstruction or the angioembolization of bleeding vessels.[5]
3. Hemodynamically stable patients who do not have hematuria and even patients with microscopic hematuria can be managed conservatively with careful monitoring.

4. Vigorous hydration to dilute gross hematuria will facilitate the avoidance of clot formation and the potential for urinary obstruction.
5. Immediate surgical exploration is suggested for patients with testicular trauma when testicular rupture is suspected.[13]
6. Surgical intervention is recommended for complicated intra- or extraperitoneal bladder injury and conservative management with Foley catheter drainage for uncomplicated extraperitoneal injuries.[18]

RETURN-TO-PLAY GUIDELINES

No specific guidelines are available to dictate the timing for a return to maximal activity after GU trauma; thus prudent decisions should take into account the severity of the injury and the required recovery period. Contact sport activities such as football should be avoided for a longer period during recovery, and specific protective padding to shield the flank regions may be indicated.

Recommendations for safe athletic participation in special populations have been controversial. For example, athletes with either a single kidney or those who have undergone a kidney transplant are intuitively considered at higher risk for the effects of GU trauma. However, the American Academy of Pediatrics recommends a "qualified yes" for participation in contact/collision sports by athletes with a single kidney.[21]

Despite this recommendation, most physicians continue to discourage participation in contact/collision sports for patients with a single kidney.[22]

Patients who have had a renal transplant are also believed to be at higher risk for trauma to the allograft because of the unique location of the allograft in the preperitoneal space. Despite the potential for increased risk of trauma to the allograft, few actual reports of such trauma have surfaced, and the medical literature does not include any reports of allograft loss due to trauma.[23,24] Given the risk of severe consequences with allograft trauma, the general consensus is that contact sports should be avoided altogether by patients who have undergone renal transplantation.[24] Individual decisions should be made for other sports that have a theoretical reduced risk of injury, such as basketball, mountain biking, and other sports.

METABOLIC PROBLEMS IN RENAL SPORTS MEDICINE

Acute Kidney Injury, Hyponatremia, and Rhabdomyolysis

Acute kidney injury (AKI) is not an infrequent occurrence in endurance sports. In one study, 83.6% of 28 ultramarathon runners developed AKI with elevations in serum creatinine levels after the race concluded, but recovery of renal function was quick, within 24 hours. Dehydration and rhabdomyolysis are notable risk factors for AKI in endurance events.[25]

Acute hyponatremia (serum sodium <135 mEq/L) can be seen when there is excessive ingestion of water in endurance sports or water drinking challenges. This can lead to severe or critical hyponatremia, leading to encephalopathy, seizures, pulmonary edema, and death.[26,27] Hyponatremia is not infrequently seen in marathon runners and triathletes and has even been reported in a training camp for rowers that lasted for several weeks.[28–30] The mechanism is dilutional, due to inadequate solutes in fluids ingested, solute loss in sweat, and inappropriate release of arginine vasopressin (antidiuretic hormone) preventing free water loss by the kidney. The therapeutic approach in symptomatic patients is an infusion of hypertonic 3% normal saline.[26]

Exertional rhabdomyolysis (ER) is relatively uncommon and often goes unrecognized. However, ER can have significant morbidity and mortality if it is severe and not managed appropriately. In such instances, inpatient hospitalization is required for intravenous hydration and serial laboratory monitoring of renal function and electrolytes. Consultation with a nephrologist should be sought in the case of athletes who have muscle enzymes that are continuing to rise. The administration of diuretics is not indicated, as this can worsen kidney function. There are no evidence-based guidelines for return to play after an episode of ER.[31]

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Dane B, Baxgter AB, Bernstein MP. Imaging genitourinary trauma. *Radiol Clin N Amer*. 2017;55(2):321–335.

Level of Evidence:
V

Summary:
Article gives great overview of types of imaging modalities best suited for patients with trauma to the genitourinary tract in appropriate clinical scenarios.

Citation:
Bjurlin MA, Fantus RJ, Fantus RJ, et al. Comparison of nonoperative and surgical management of renal trauma: Can we predict when nonoperative management fails? *J Trauma Acute Care Surg*. 2017;88(2):356–362.

Level of Evidence:
II

Summary:
A retrospective cohort study analyzing risk factors that determine the success of salvage and failure of nonoperative management of renal trauma.

Citation:
Chiron P, Hornez E, Boddaert GM, et al. Grade IV renal trauma management. A revision of the AAST renal injury grading scale is mandatory. *Eur J Trauma Emerg Surg*. 2016;42(2):237–241.

Level of Evidence:
I

Summary:
Systematic review of existing literature on outcomes of Grade IV renal trauma based on imaging and risk factor stratification with recommendations for management.

REFERENCES

1. Santucci RA, McAninch JW, Safir M, et al. Validation of the American Association for the Surgery of Trauma organ injury severity scale for the kidney. *J Trauma*. 2001;50:195–200.
2. Bagga HS, Tasian GE, Fisher PB, et al. Product related adult genitourinary injuries treated at emergency departments in the United States from 2002 to 2010. *J Urol*. 2013;189:1362–1368.
3. Bagga HS, Fisher PB, Tasian GE, et al. Sports-related genitourinary injuries presenting to United States emergency departments. *Urology*. 2015;85:239–244.
4. Grinsell MM, Butz K, Gurka MJ, et al. Sport-related kidney injury among high school athletes. *Pediatrics*. 2012.
5. Santucci RA, Wessells H, Bartsch G, et al. Evaluation and management of renal injuries: consensus statement of the renal trauma subcommittee. *BJU Int*. 2004;93:937–954.
6. Emanuel B, Weiss H, Gollin P. Renal trauma in children. *J Trauma*. 1977;17:275–278.
7. Bjurlin MA, Zhao LC, Goble SM, et al. Bicycle-related genitourinary injuries. *Urology*. 2011;78:1187–1190.
8. McAleer IM, Kaplan GW, LoSasso BE. Renal and testis injuries in team sports. *J Urol*. 2002;168:1805–1807.
9. Haas CA, Brown SL, Spirnak JP. Penile fracture and testicular rupture. *World J Urol*. 1999;17:101–106.
10. Schmidlin FR, Iselin CE, Naimi A, et al. The higher injury risk of abnormal kidneys in blunt renal trauma. *Scand J Urol Nephrol*. 1998;32:388–392.
11. Chopra P, St-Vil D, Yazbeck S. Blunt renal trauma-blessing in disguise? *J Pediatr Surg*. 2002;37:779–782.
12. Onen A, Kaya M, Cigdem MK, et al. Blunt renal trauma in children with previously undiagnosed pre-existing renal lesions and guidelines for effective initial management of kidney injury. *BJU Int*. 2002;89:936–941.
13. Styn NR, Wan J. Urologic sports injuries in children. *Curr Urol Rep*. 2010;11:114–121.
14. Dat A, Chew LY, Hanegbi U. Beware of the "ascites" patient: delayed presentation of traumatic intraperitoneal bladder rupture. *Urology Case reports*. 2016;4:11–13.
15. Nicolaisen GS, McAninch JW, Marshall GA, et al. Renal trauma: re-evaluation of the indications for radiographic assessment. *J Urol*. 1985;133:183–187.
16. Miller KS, McAninch JW. Radiographic assessment of renal trauma: our 15-year experience. *J Urol*. 1995;154:352–355.
17. Bretan PN Jr, McAninch JW, Federle MP, et al. Computerized tomographic staging of renal trauma: 85 consecutive cases. *J Urol*. 1986;136:561–565.
18. Quagliano PV, Delair SM, Malhotra AK. Diagnosis of blunt bladder injury: a prospective comparative study of computed tomography cystography and conventional retrograde cystography. *J of Trauma: Inj, Infect and Crit Care*. 2006;61:410–422.
19. Holmes FC, Hunt JJ, Sevier TL. Renal injury in sport. *Curr Sports Med Rep*. 2003;2:103–109.
20. Kraus SE, Siroky MB, Babayan RK, et al. Hematuria and the use of non-steroidal anti-inflammatory drugs. *J Urol*. 1984;132:288–290.
21. Rice SG. Medical conditions affecting sports participation. *Pediatrics*. 2008;121:841–848.
22. Sharp DS, Ross JH, Kay R. Attitudes of pediatric urologists regarding sports participation by children with a solitary kidney. *J Urol*. 2002;168:1811–1814, discussion 1815.
23. Ahmed S, Batiuk TD. Broken Kidney: traumatic fracture of a renal allograft. *Am J Kidney Dis*. 2001;37:E33.
24. Heffernan A, Gill D. Sporting activity following kidney transplantation. *Pediatr Nephrol*. 1998;12:447–448.
25. Hou S-K, Chiu Y-H, Tsai Y-F, et al. Clinical impact of speed variability to identify ultramarathon runners at risk for acute kidney injury. *PLoS ONE*. 2015;10(7):e0133146. doi:10.1371/journal.pone.0133146.
26. Ayus JC, Varon J, Arieff AI. Hyponatremia, cerebral edema, and noncardiogenic pulmonary edema in marathon runners. *Ann Intern Med*. 2000;132:711–714. doi:10.7326/0003-4819-132-9-200005020-0000.
27. Siegel A. Fatal water intoxication and cardiac arrest in runners during marathons: prevention and treatment based on validated clinical paradigms. *Am J of Med*. 2015;128:1170–1175.
28. Almond CSD, Shin AY, Fortescue EB, et al. Hyponatremia among Runners in the Boston Marathon. *N Engl J Med*. 2005;352:1550–1556.
29. Danz M, Pöttgen K, Tönjes PM, et al. Hyponatremia among Triathletes in the Ironman European Championship. *N Engl J Med*. 2016;374:997–999.
30. Mayer CU, Tredd G, Fenske WK, et al. High incidence of hyponatremia in rowers during a four-week training camp. *Amer J of Med*. 2015;128:1144–1151.
31. Tietze DC, Borchers J. Exertional rhabdomyolysis in the athlete. A clinical review. *Sports Health*. 2014;4:336–339.